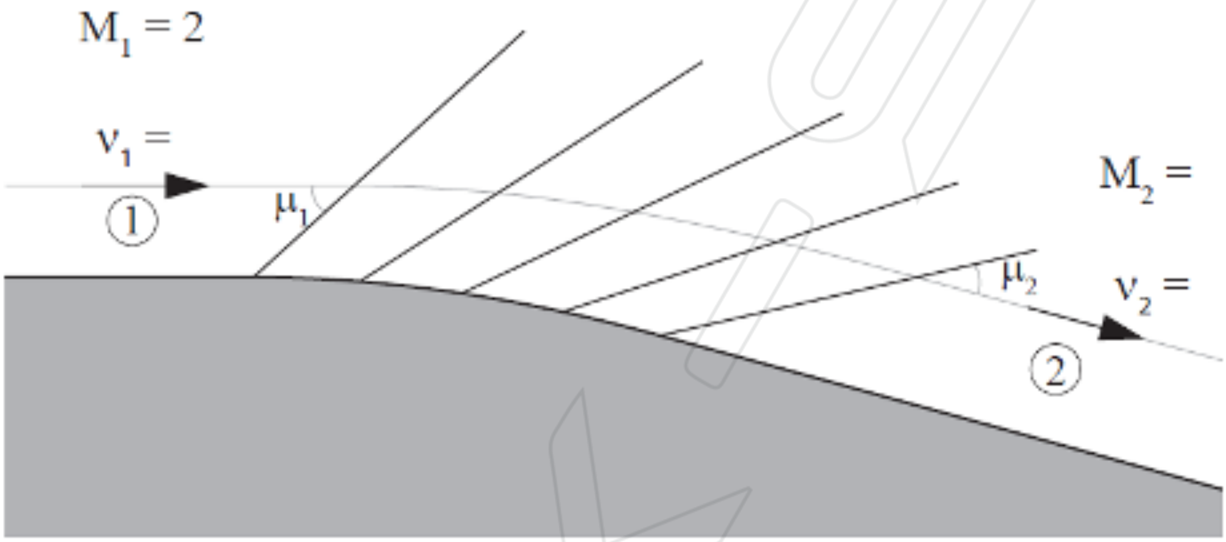
AERO50001 Aerodynamics 2 Problem Set #5

Extra practice: Expansion fans, shock-expansion theory

5.1 Expansion fan

★★☆ 15-25 mins

Determine the following quantities downstream of the continuous $10°$ convex corner shown below, with the flow upstream of the corner being at Mach 2:



(a) The Mach number, M_2.

(b) The pressure, p_2, as a function of the upstream pressure p_1.

(c) Find the angles μ_1 and μ_2 that the forward and rearward Mach lines form with respect to their local flow.

(a) Left-running : $\theta_1 + \nu(M_1) = \theta_2 + \nu(M_2)$

$\therefore \nu(M_2) = \theta_1 + \nu(M_1) - \theta_2 = 0 + 26.38° - (-10°) = 36.38°$

$\therefore M_2 = 2.39$

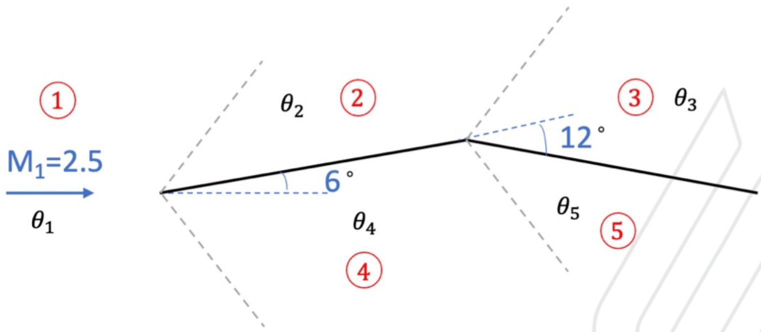
(b) $\frac{p_2}{p_1} = \frac{p_2}{p_{02}}\frac{p_{02}}{p_{01}}\frac{p_{01}}{p_1} = \frac{p_2}{p_{02}}\frac{p_{01}}{p_1} = \frac{7.8244}{14.393} = 0.5436$

(c) $\mu_1 = 30°$, $\mu_2 = 24.73°$

5.2 Cranked aerofoil

★★☆ 15-30 mins

Use shock-expansion theory to find the pressure on every surface of the following cranked aerofoil in a Mach 2.5 flow (relative to p_1).



At ②, $M_2 = 2.251$

$\therefore \frac{p_2}{p_1} = 1.468$

At ④, $\theta_1 - \nu(M_1) = \theta_2 - \nu(M_4)$

$\therefore \nu(M_4) = \theta_2 - \theta_1 + \nu(M_1) = 6° - 0° + 39.12° = 45.12°$

$\therefore M_4 = 2.77, \ \frac{p_4}{p_1} = \frac{p_4}{p_{04}} \frac{p_{04}}{p_{01}} \frac{p_{01}}{p_1} = \frac{17.0859}{25.922} = 0.6591$

At ③, $\theta_2 + \nu(M_2) = \theta_3 + \nu(M_3)$

$\therefore \nu(M_3) = \theta_2 + \nu(M_2) - \theta_3 = 6° + 33.02° - (-6°) = 45.02°$

$\therefore M_3 = 2.77, \ \frac{p_3}{p_1} = \frac{p_3}{p_{03}} \frac{p_{03}}{p_{02}} \frac{p_{02}}{p_2} \frac{p_2}{p_1} = \frac{11.5812}{25.922} \cdot 1.468 = 0.4193$

At ⑤, $\theta_4 - \nu(M_4) = \theta_5 - \nu(M_5)$ ($\theta_{local} = 12°$)

$\therefore \nu(M_5) = \theta_5 - \theta_4 + \nu(M_4) = -6° - 6° + 45.12° = 33.12°$

$\therefore M_5 = 2.2275, \ \frac{p_5}{p_1} = \frac{p_5}{p_4} \frac{p_4}{p_1} = 2.223 \cdot 0.6591 = 1.465$